

ORGANIC CHEMISTRY

January 2026

Dr. Sardar N. Ahmed

ORGANIC CHEMISTRY

ORGANIC CHEMISTRY

Is the study of carbon containing compounds. The element carbon has a special role in chemistry because it bonds with other carbon atoms to give many compounds, that we find organic compounds in many common products, we use so many of molecules. The variety of molecules is so great such as gasoline, medicine, shampoos, plastic bottles, and perfumes.

The food we eat is composed of different organic compounds that supply us with fuel for energy and the carbon atoms needed to build and repair the cells of our bodies. Large organic molecules make up the proteins in hair and skin, and the DNA in cell nuclei.

At the beginning of the nineteenth century, scientists classified chemical compounds as inorganic and organic.

An inorganic compound was a substance that was composed of minerals, and an organic compound was a substance that comes from an organism.

Thus the use of the word “vital force” which could only found in living cells, was required to synthesize an organic compound.

This perception was shown to be incorrect in 1828, when the German chemist Friedrich Wohler synthesized urea, a product of protein metabolism, by heating an inorganic compound Ammonium cyanate.

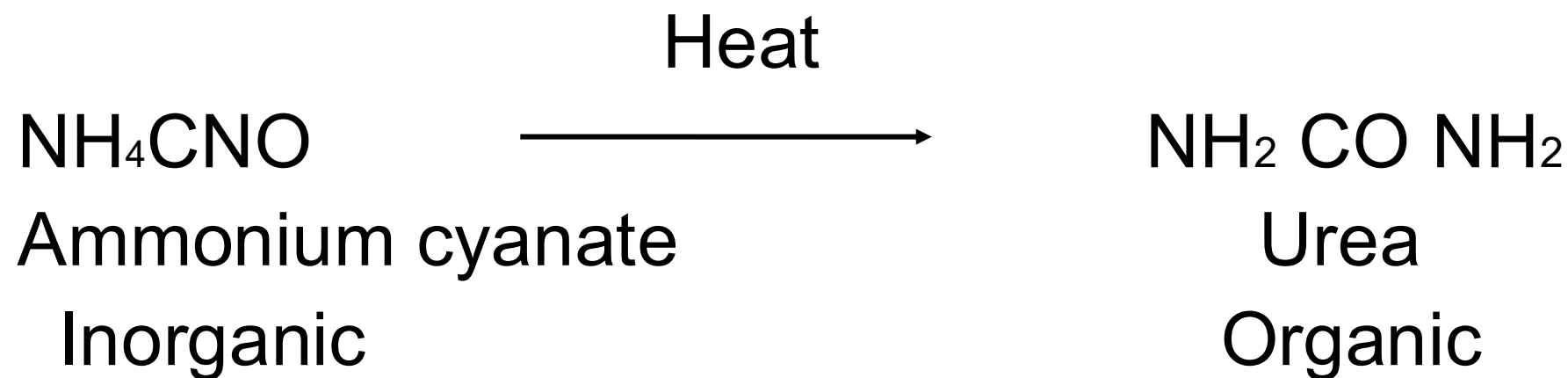
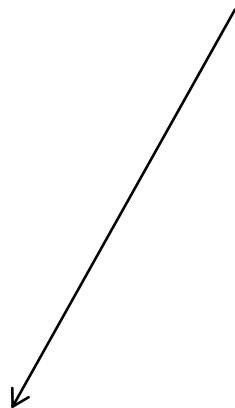


Table (1): A comparison of properties of organic and inorganic compounds

ORGANIC COMPOUNDS	INORGANIC COMPOUNDS
○ Bonding is almost entirely covalent ○ May be gases, liquid, solid ○ Most are insoluble in water ○ Most are soluble in organic solvents such as diethyl ether, toluene, benzene ○ Aqueous solutions do not conduct electricity ○ Low boiling and melting points ○ Almost all burn ○ Reactions are usually slow	○ Most have ionic bonds ○ Most are solids ○ Most are soluble in water ○ Almost all are insoluble in organic solvents. ○ Aqueous solutions forms ions that conduct electricity ○ High boiling and melting points ○ Very few burn ○ Reactions are often very fast

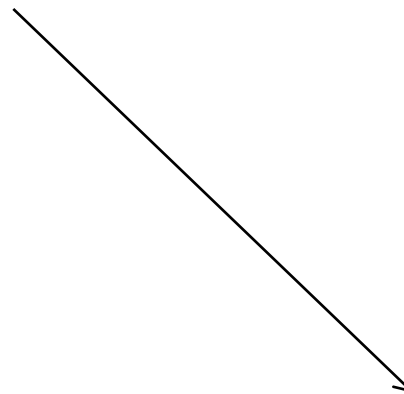
CLASSIFICATION OF ORGANIC COMPOUNDS

ORGANIC COMPOUNDS



Aliphatic hydrocarbons

Types of aliphatic



Aromatic hydrocarbons

Hydrocarbons

Hydrocarbons are organic compounds composed of carbon and hydrogen elements only, subdivided to;

- Alkanes all C-C bonds are saturated $R-CH_2CH_3$

General formula C_nH_{2n+2}

- Alkenes contain at least one double bond $R-CH=CH_2$

General formula C_nH_{2n}

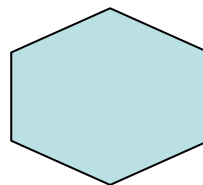
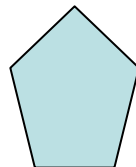
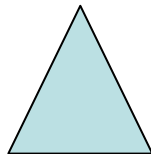
- Alkynes contain at least one triple bond $RC\equiv CH$

General formula C_nH_{2n-2}

Cycloalkanes are cyclic alkanes when two ends bind together after remove two hydrogen atoms, general formula C_nH_{2n}

Examples;

Cyclopropane, cyclobutane, cyclopentane, cyclohexane, ext....



Complete combustion

Alkane + oxygen $\longrightarrow$ CO₂ + WATER

Incomplete combustion

Alkane + low oxygen $\longrightarrow$ CO + WATER

Low oxygen produces more CO

Carbon monoxide is toxic, no odor, colorless, causes victims will
Feel dizzy, loss consciousness, and if not removed from the CO,
they can die

Incomplete combustion often takes place inside a car engine due to a
limited amount of oxygen present.

Alkenes

Ethene

$\text{CH}_2=\text{CH}_2$ Plant growth hormone

ALPHA- Farnensene

Composed of 15- carbon atoms and four double bonds

α : 3,7,11-trimethyl-1,3,6,10-dodecatetraene

Found in apple responsible for odor and color of green apple.

Aromatic compounds

Aromatic compounds

Aromatic compounds:

Are chemical compounds with six member rings containing alternative single and double bonds between carbon atoms.

Benzene C₆H₆

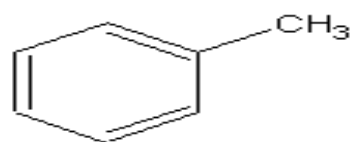
Benzene structure is the basis of other aromatic compounds.

Benzene

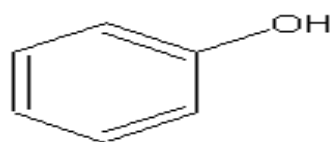
Is toxic when taken internally, it is considered to be carcinogenic.

Naming aromatic compounds;

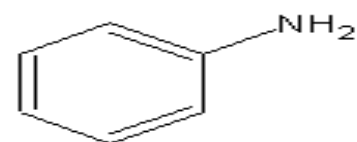
Aromatic compounds that contain a benzene ring with a substituent are usually named as benzene derivatives. However, many of these compounds have been important in chemistry for many years and still use their common names.



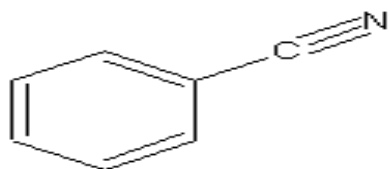
toluene



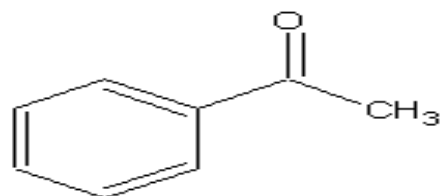
phenol



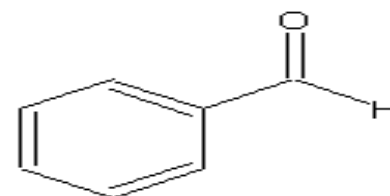
aniline



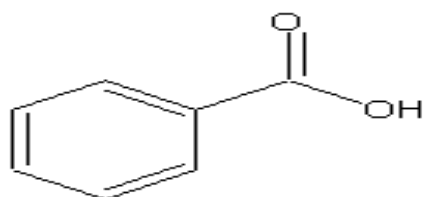
benzonitrile



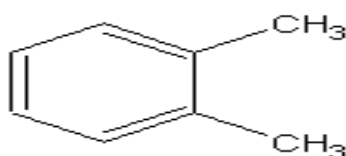
acetophenone



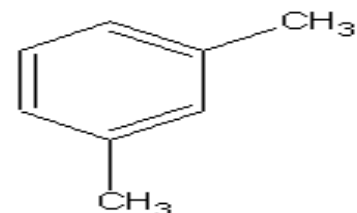
benzaldehyde



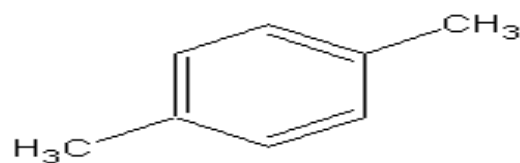
benzoic acid



o-xylene



m-xylene



p-xylene

Substitution reactions of aromatic compounds;

The most common reaction of the benzene ring involves substituting one or more of the ring hydrogen's by other groups or atoms.

1-Halogenation

2-Nitration

3-Sulfonation

4-Friedel-Crafts Alkylation

5-Friedel-Crafts Acylation

Aromatic compounds in health and medicine;

Aromatic compounds are common in nature and in medicine.

-Toluene-($C_6H_5-CH_3$) is used as;

1- Preservative of urine samples.

2- Reactant to make drugs, dyes and explosives such as TNT (Tri nitro toluene).

The benzene ring is found in some amino acids (the building blocks of proteins).

In pain relievers, such as,
Aspirin (Acetyl salicylic acid)
and
Acetaminophen (Paracetol).

Poly cyclic aromatic hydrocarbon's (PCAHC'S).

Large organic compounds known as Poly cyclic aromatic hydrocarbon's (PCAHC'S). They are formed by fusing together two or more benzene rings edge-to-edge.

Examples

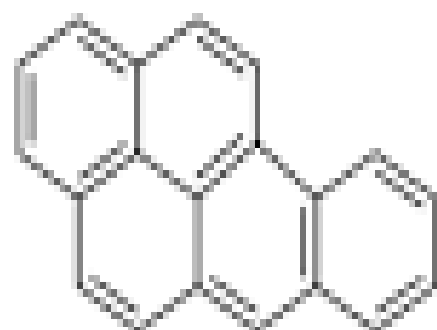
Naphthalene, Anthracene, Phenanthrene.

When a polycyclic compound contains phenanthrene, it may act as a carcinogen, a substance known to cause cancer. For example, some aromatic compounds in cigarette smoke cause cancer, as seen in the lung tissues of a heavy smoker.

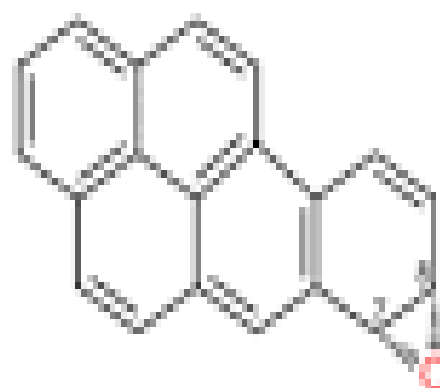
Benzapyrene a product of combustion has been identified in coal tar, tobacco, barbecued meats, and car exhaust.

Benzapyrene causes cancer in the following way

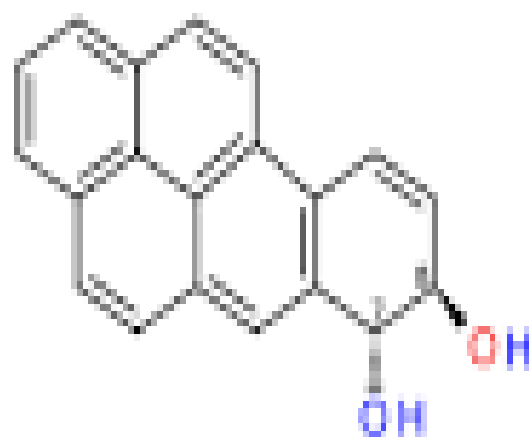
Once it is absorbed or ingested, the body attempts to convert it into a more water soluble compound that can be excreted easily. By a series of enzyme-catalyzed reactions, Benzapyrene is transformed into a diol (two-OH groups) epoxide (a three membered ring, one atom of which is oxygen). This compound can bind to DNA by reacting with one of the amino groups, thereby altering the structure of DNA and producing a cancer-causing mutation.



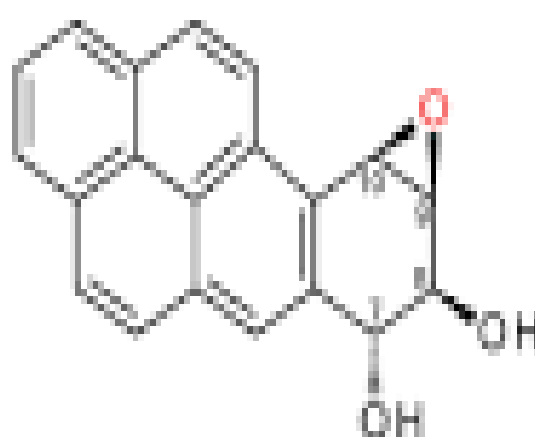
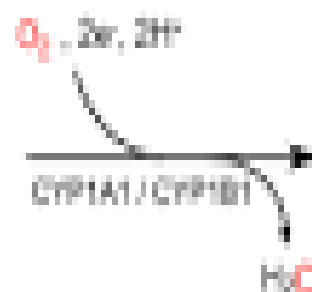
benzo[a]pyrene



(+)-benzo[a]pyrene-7,8-epoxide



(-)-benzo[a]pyrene-7,8-dihydrodiol



(+)-benzo[a]pyrene-7,8-dihydrodiol-9,10-epoxide

Heterocyclic aromatic compounds

Heterocyclic aromatic compounds

Compounds are those having at least one atom other than carbon as part of the structure of the aromatic ring.

EXAMPLES:

pyridine:

Pyridine is a basic heterocyclic organic compound with the chemical formula C_5H_5N . It is structurally related to benzene, with one C-H group replaced by a nitrogen atom. The pyridine ring occurs in many important compounds

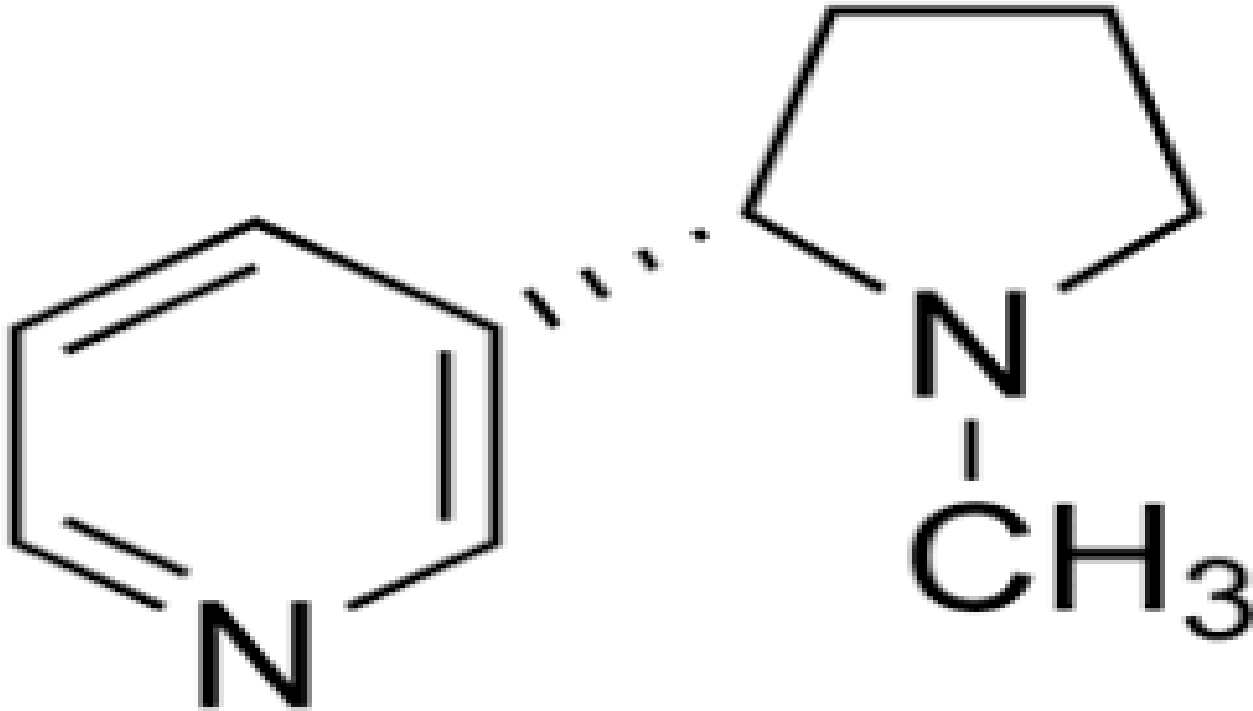
Pyridine

—



Pyridine

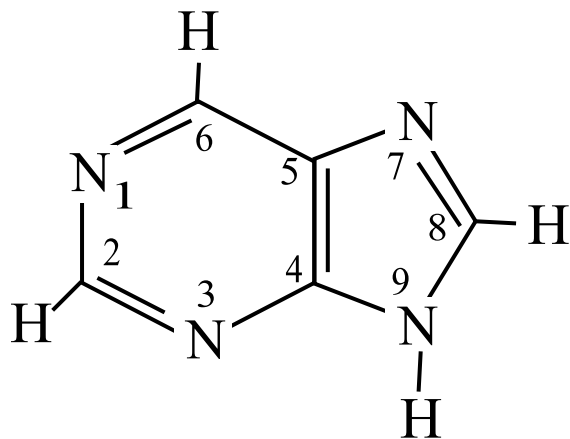
The pyridine ring is found in nicotine.
The addictive compound in tobacco



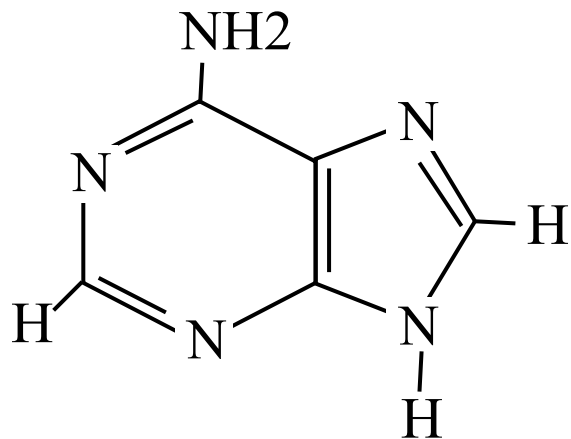
Purines and pyrimidines:

Are components of DNA &RNA;

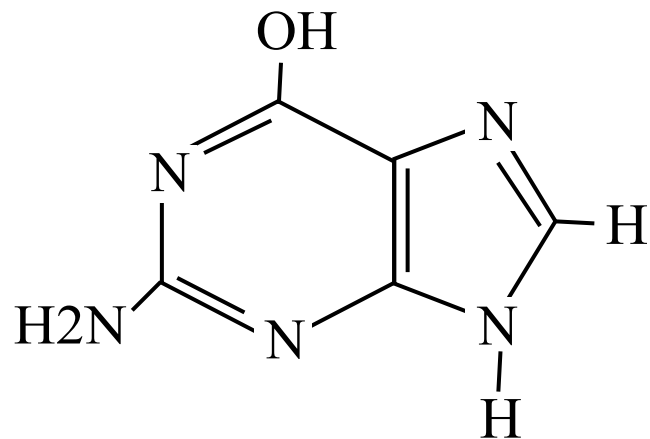
They are molecules responsible for storing and expressing the genetic information of an organism.



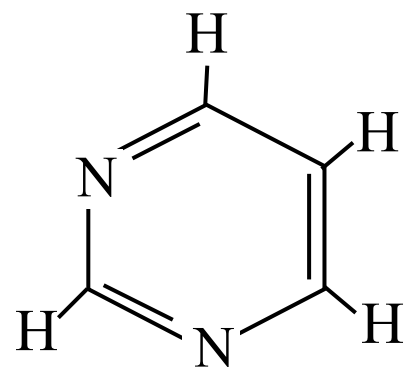
Purine (Parent compound)



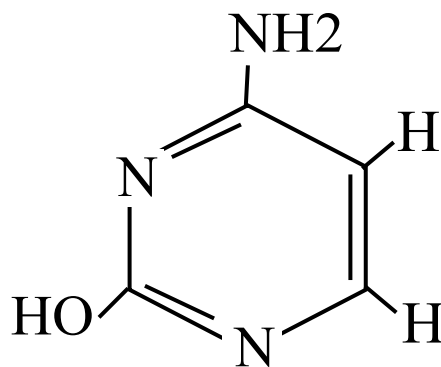
Adenine (A)



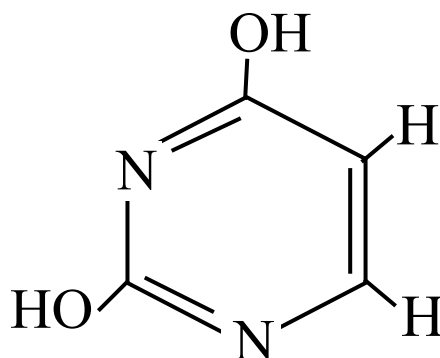
Guanine (G)



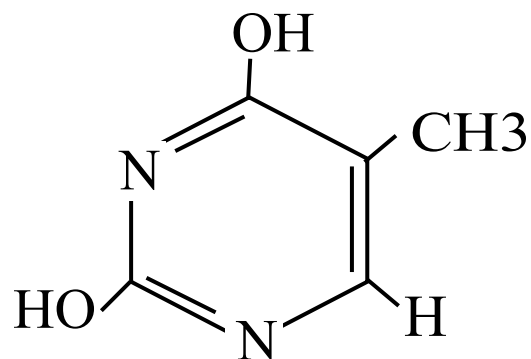
Pyrimidine
(Parent compound)



cytosine ©



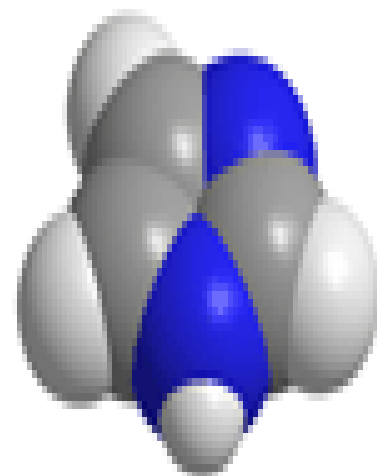
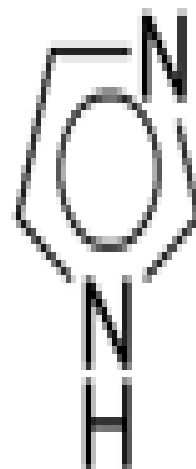
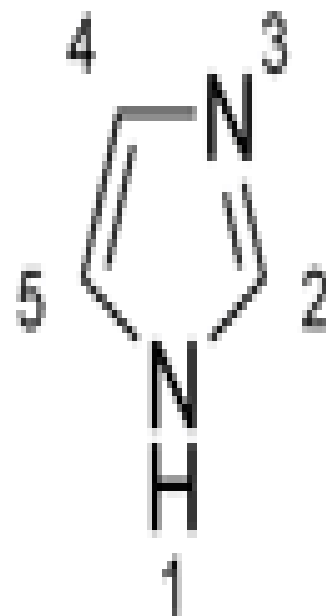
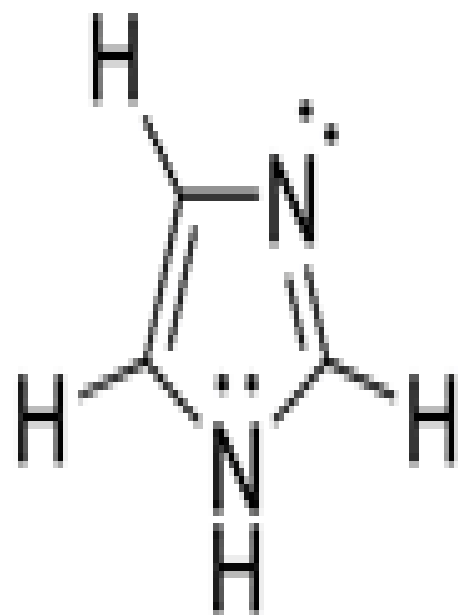
uracil (U)



thymine (T)

-Imidazole

The imidazole ring is a component of cimetidine, a drug used in the treatment of stomach ulcers, and 4-Nitroimidazole is an imidazole derivatives. Its antibiotics that have been used to combat anaerobic bacterial and parasitic infections. perhaps the most common example is metronidazole (Flagyl)

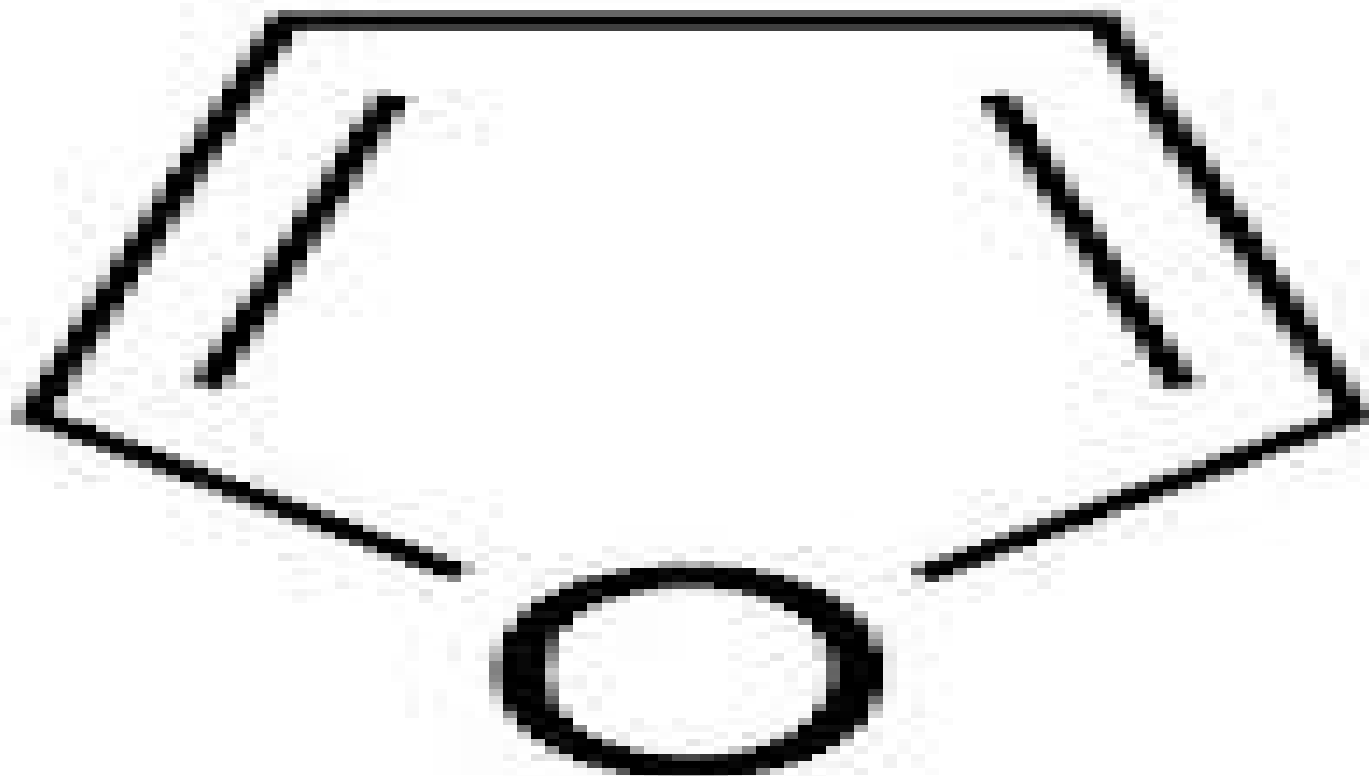


Furan:

It's a heterocyclic organic compound, consisting of a five-membered aromatic ring with four carbon atoms and one oxygen.

Furan is a colorless, flammable, non polar solvent, highly volatile liquid with a boiling point close to room temperature. It is toxic and may be carcinogenic

Furan



Pyrrole

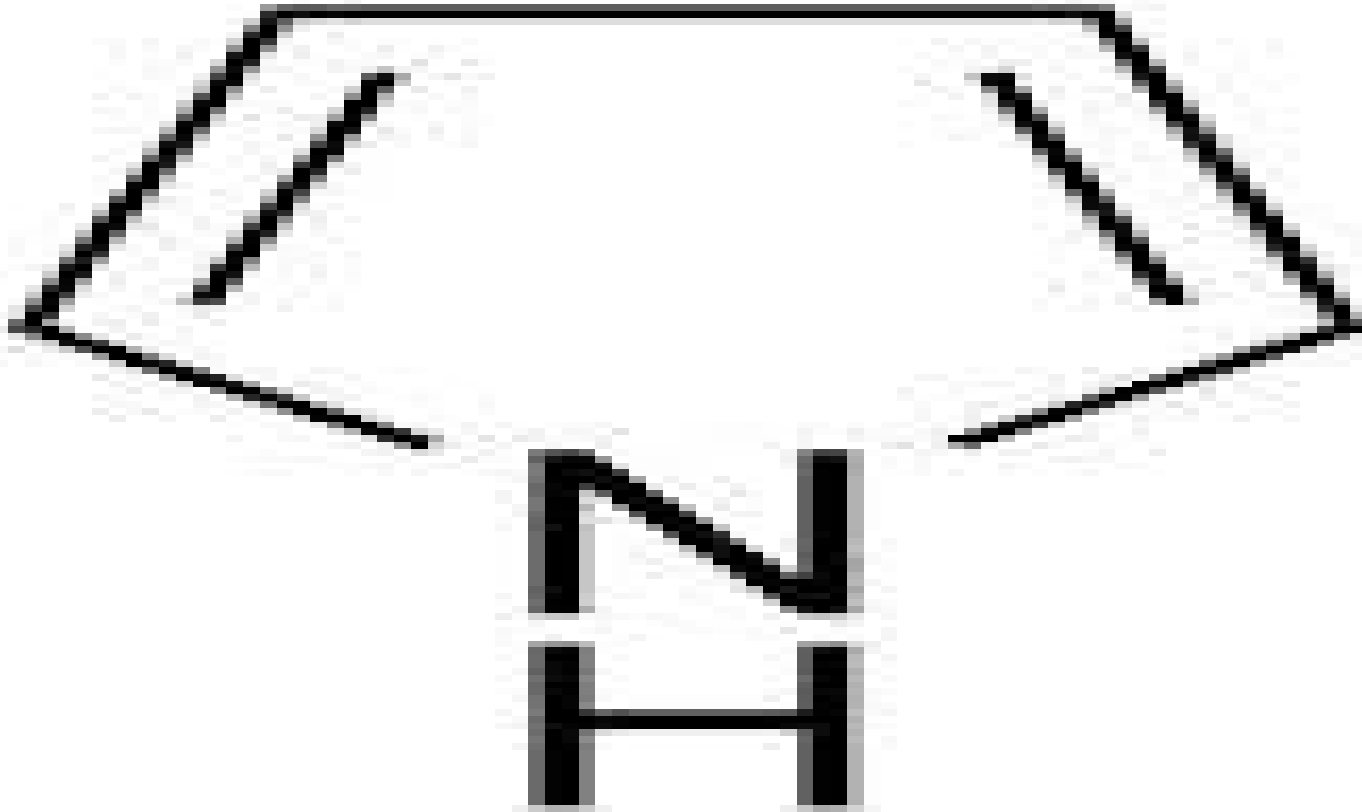
The pyrrole ring is a component of porphyrine ring found in hemoglobin and chlorophyll.

Pyrrole:

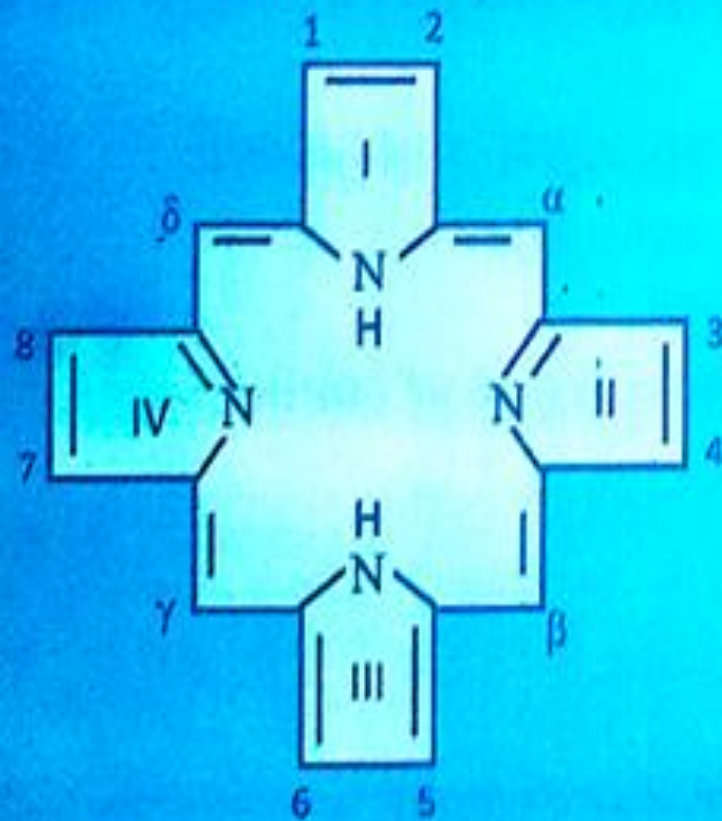
Is a heterocyclic aromatic organic compound, a five-membered ring with the formula C_4H_4NH . It is a colorless volatile liquid that darkens readily upon exposure to air.

Pyrrole

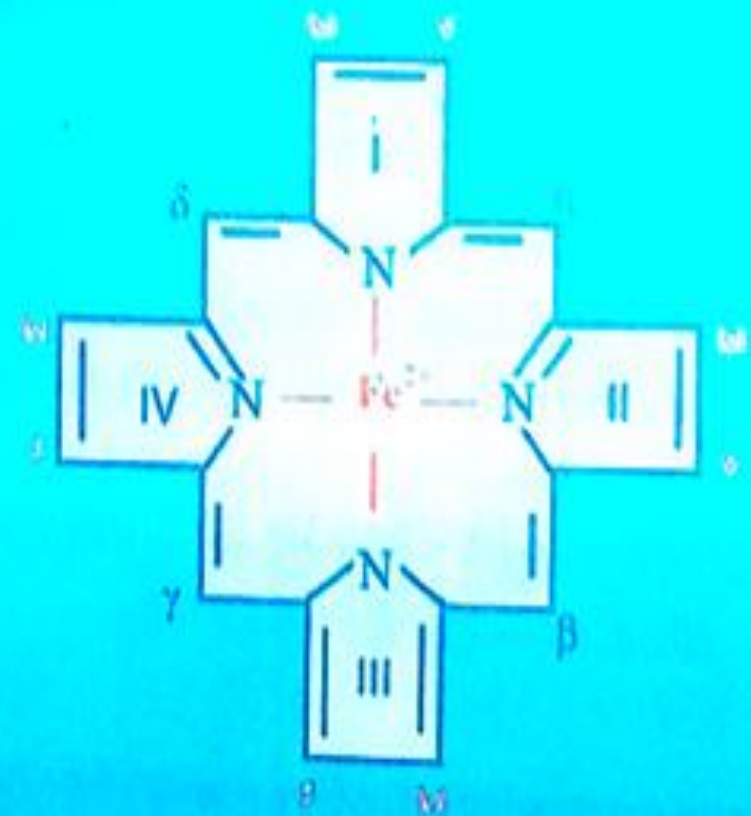
—



Porphin Ring



Heme [Ferrous Protoporphyrin IX]



M= methyl group (CH_3), V= vinyl group ($\text{CH}=\text{CH}_2$)
 P= propionate ($\text{CH}_2\text{-CH}_2\text{-COO}^-$)